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Final Report

My goal for the baseline PPA was to finish the flow no matter how bad the metrics turned out to be. However, due to there being violations that I could not fix, I could not reach that goal. This also applies to the improved versions, so I will analyze all of their metrics while acknowledging their failures in this report.

BASE DESIGN:

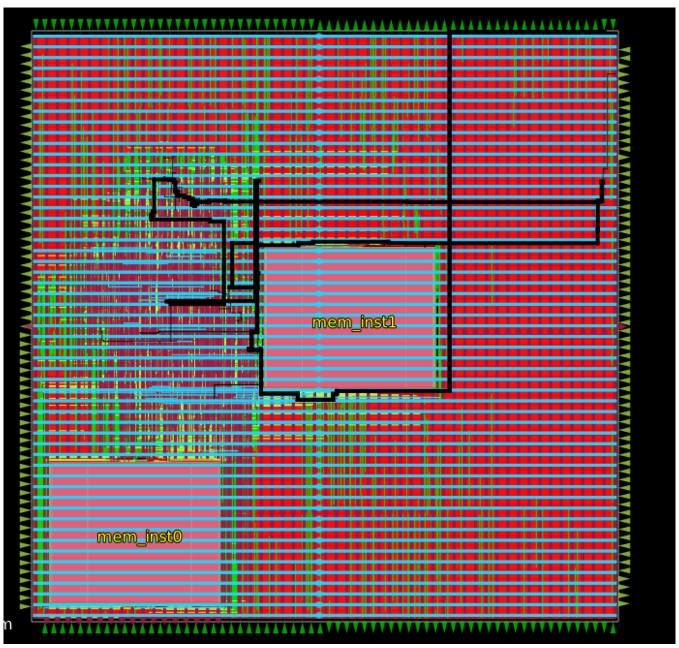
For my base design, I decided to add as many optimizations as I could to ensure that the result had no violations and ran properly. First, I made sure to give adequate space to my memory macros (placed at [50,50] and [650,650] respectively). This ensures that there are little to no routing issues in regards to the placement of the macros while still keeping them close to each other. Second, I switched on many optimization steps like the post grt resizer timing and the post grt design repair steps. This helped reduce the number of violations that the flow was generating. Third, I introduced slack margins to help optimize the design. It does this by aiming for a positive slack rather than just zero slack (which I set to 0.2). Lastly, I turned on the heuristic diode insertion to add the antennas to the memory macros (and gave it 30 iterations just to make sure there were not any errors). This got rid of the majority of the antenna errors. There are many violations that still exist even though I ran the optimizer steps, so I will talk about those in the failures section.

POWER:

Power wise, the baseline PPA uses 0.016116395592689514 W in total, with the voltage being split into internal, switching, and leakage. The internal voltage was 0.005237452220171 69W, the switching voltage was 0.010840564966201782 W, and the leakage was 3.837839904008433e-05 W. The leakage was much lower in comparison to the other statistics with the switching voltage being the highest. Another interesting metric to look at was the design_powergrid__drop__worst__net:vssd1__corner:nom_tt_025C_1v80 where the value was 0.000347809. This was about 2 percent of the total voltage, and considering how many nets there are in this circuit, this value is pretty huge.

TIMING:

Here is an image of the critical path:



This is the path with the least amount of slack (89.041). This path is very long and connects to multiple areas. It also passes around the memory macro and into the highly congested area. Clearly, it has many long wires. The fanout also seems to be pretty high considering it splits into at least 5 different locations. Checking the fanout, the value was 30.

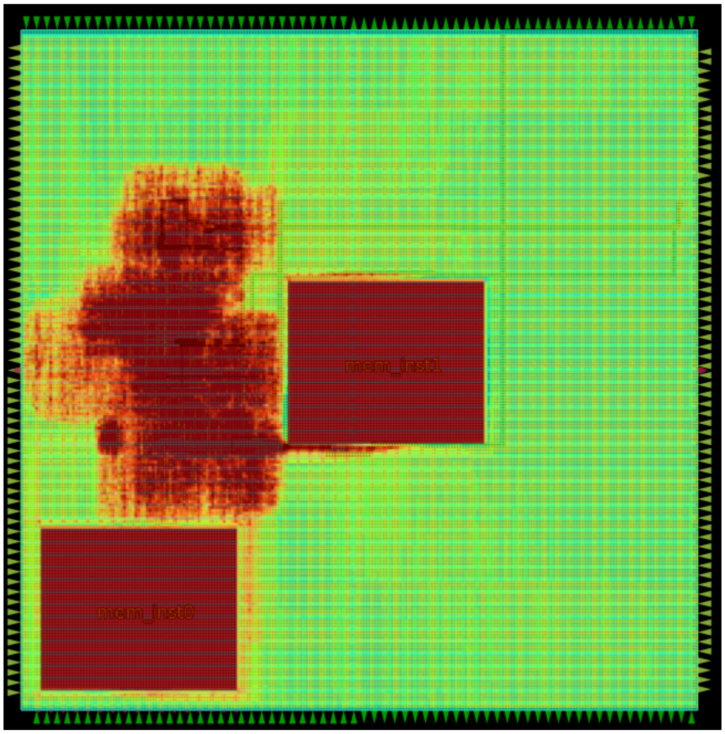
The maximum clock skew in this design was 3.537. The critical path however did not have such a high clock skew with around 3.406.

FLOORPLAN:

In my base design, I utilized around 20 percent of the core area for placement. My thought process behind this was that I wanted to keep the baseline as free from congestion as possible, as a very low density of the objects placed allowed for very easy routing. However, the downside is that the resulting area is very large. As you can see, compared to my improved version (which has a core utilization of 35), the area of my baseline is much larger (641643 vs. 609232). This is an okay margin for the increase of core utilization.

As for the wirelength in my design, the total wirelength of the design was (). This value might be smaller than others because I set the maximum wirelength to be 200 microns. This may have caused more congestion in the process, but with the low core utilization, it should have been fine. The tradeoff of course was less wirelength and less overall die size.

Speaking of congestion, most of the congestion was focused in the area to the left of mem1 instance and above the mem0 instance. This makes sense since most operations need to access these macros to work. Here is a heat map of the congestion:



IMPROVEMENTS:

The two areas I wanted to improve were to significantly shrink the area of the circuit and reduce the clock period significantly as well to increase the performance. My first improvement was to increase the core utilization since this directly correlates with the size of the die. The more core that is utilized, the more dense the circuit is and the lower the die area becomes. I increased the core utilization from 20 to 35 which led to an okay size reduction. The size reduction was about 5-6 percent from the increase of core utilization from 20 to 35. Shockingly, the main change was not the size. It was the power usage. The power usage went from 0.016116395592689514 W to 0.012833796441555023W which is around a 20 percent decrease in power.

The second improvement was an attempt to lower the clock period and make the circuit much faster. It was a success with the hold tns going from -74.81307518574864ns to -4.173876473696207ns. This is a very large decrease in time, nearly 20 times. The setup tns decreased a little though, going from 88.69116891943878ns to 69.3364532081505ns. On the contrary, the power consumption was back to the baseline (0.012833796441555023W to 0.016929905861616135W) but the size stayed around the same.

The second improvement showed that improving one metric hurt another one of the metrics. In this case, it increased the speed of the design but it hurt the power consumption. In my opinion, this tradeoff was worth it since the performance was much better than the baseline while still using a similar amount of power.

FAILURES:

There were many failures in this assignment. First, while the design did pass DRC and LVS, there were multiple violations. Specifically, however I tried to fix the hold time violations I would receive very questionable errors, like the max_transition being set to 0.04ns even though if you look in my design, I set it to 1.5 ns. Second, the improvements were a bit lacking. I thought that the jump from 20 to 25 would bring at least a 40 percent reduction in the die area. I was wrong. Also, looking at the baseline PPA, there seems to be **A LOT** of space that is not being properly utilized. Specifically, the congestion was only focused in one area, which really hurt the design.